# MARCEL

8-bit looper that slowly forgets

BASIC HUMAN TASTES

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Marcel records into a small 8-bit memory read by a clock you can bend. Slow the clock and the loop gets longer, lower and grittier; speed it up and it gets short, clean and high.

And Marcel forgets. ERODE wears the loop down a little more on every pass and TONE darkens or thins it, until only a ghost of the sound is left. A ring at the top shows the memory as it fades.

## AT A GLANCE

- 16,384 cells of 8-bit memory, clocked from 1 to 48 kHz: loops from 16 s down to 0.34 s
- Changing the speed bends the pitch of what is already recorded
- ERODE and TONE act on every pass, even on a frozen loop
- Reverse, reset, freeze, an end-of-loop trigger and a clock output
- 12 HP

## CONTROLS

| | |
|---|---|
| Ring | Each spoke is the loudness of one slice of the memory; the white dot is the play head. Gold while recording, cream while frozen. |
| SPEED | Memory clock, from 1 to 48 kHz. The loop lasts 16,384 clock ticks: 16 s fully left, 0.34 s fully right (1.6 s at the start). Turning it while sound is in the memory changes its pitch. |
| INPUT | Level of the incoming sound written into the memory. |
| FEEDBACK | How much of the loop is written back on each pass, up to 110 %. Above 100 % the loop grows until it saturates in 8 bits. |
| ERODE | How much the memory forgets on every pass: a little volume, less resolution (from 8 bits down to 4), slices that fade to silence and soft crackles. Fully right, a frozen loop has nearly gone after 4 passes; around 30 % it takes about a dozen. |
| PITCH | Amount of the PITCH input. Fully right, 1 V per octave on the clock; left of centre inverts it. |
| TONE | A filter inside the loop. Centre: no change. Left: each pass a little darker. Right: each pass a little thinner. Its cutoff follows SPEED. |
| FREEZE | Lit: nothing new is recorded. ERODE and TONE keep working, so a frozen loop still fades unless ERODE is at zero and TONE at the centre. |

## INPUTS

| | |
|---|---|
| IN | Sound to record (±5 V). |
| PITCH | Bends the memory clock, scaled by the PITCH knob. |
| ERODE | Added to the ERODE knob; 10 V = 100 %. |
| REVERSE | Plays and records backwards while the gate is high. The light next to it shows it. |
| RESET | A trigger sends the play head back to the start of the loop. |
| TRIG | Each trigger turns FREEZE on or off. |
| INPUT OPEN / CLOSE | Two level controls for the input. A rising CV (0 to 10 V) opens OPEN and closes CLOSE. Unpatched, they let everything through. |
| FEEDBACK OPEN / CLOSE | The same for the feedback. |

## OUTPUTS

| | |
|---|---|
| END | A 10 V trigger each time the play head completes the loop, frozen or not. The light flashes. |
| CLOCK | Square wave at the memory clock divided by 32 (±5 V): a tone that follows SPEED. |
| OUT | The memory, ±5 V. Each value is held until the next clock tick, with no smoothing: that is where the grain comes from. |

## HOW IT WORKS

**One cell per tick**

On each clock tick Marcel reads one cell, then writes the input plus the feedback back into it and moves on. The loop therefore always lasts exactly 16,384 ticks, and recorded sound plays back faster or slower as the clock changes.

### How forgetting works

Whatever comes round again passes through TONE and ERODE before it is written back. With ERODE above zero it loses a little volume and resolution, so quiet parts disappear first; now and then a slice fades to silence or a small crackle appears. The ring makes it visible: watch it hollow out.

## PATCH IDEAS

### 1. A voice that crumbles

Microphone into IN, FEEDBACK 90 %. Record a phrase, press FREEZE, then raise ERODE to around 50 % and let it fall apart.

### 2. Varispeed tape

A slow triangle LFO into PITCH, PITCH knob around 30 %, FEEDBACK 70 %.

### 3. Back and forth

A slow square LFO into REVERSE.

### 4. Record one pass, play the next

END into TRIG: the loop freezes after one pass and opens again on the next.

### 5. Hear the loop length

END into the TRIG of an Ernest: one hit per pass.

## SPECIFICATIONS

| | |
|---|---|
| Width | 12 HP |
| Memory | 16,384 × 8 bits |
| Clock | 1 to 48 kHz, plus CV |
| Loop length | 0.34 to 16 s |
| Levels | Audio ±5 V. END 10 V, 1 ms. Gates and triggers about 1 V. |